Loop 1,

Agenda:

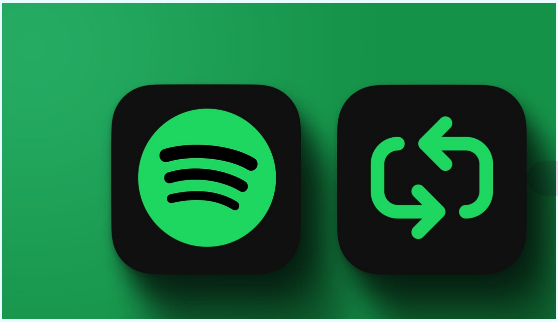
1. Need for loops
2. While loops
3. Problem solving
4. End, sep in Print
5. For loop → Intro
   - ↳ List
   - ↳ Range
   - ↳ Jump in Range
6. Problems → for loop

what are loops ?

Repeatedly execute a block of statement until a given condition is True

→ On the loop ! → → – entry / No entry ?

(M)



→ loop

→ until you get bored, play the same song again & again !

↓

while song doesn't seem to boring :

Keep playing !

until you enter password correctly : (3 attempt)

Keep trying

→

Loops

for — while

while → until

Jab Tak

until you are alive:

Do something

what is while loop ?

→ Iterate over the block of code as long as condition is True

→ when condition becomes false, codes stop !

" we use while loop when we don't know no of times we have iterate beforehand "

↓

for / while

why use while loop ?

1. Automation of Repeated Task
2. Indefinite iteration
3. Reduce code complexity

Lets say you want to hack this lock ?

→ open !

→ 0 to 9 → $10 \times 10 \times 10$

→ 0 to 9 ↓

→ 0 to 9 1000

Total combination possible →

$10 \times 10 \times 10 = 1000$

without loop :

↓

Manually check condition 1000 times

i.e

1000 → if else condition

until

↓

→ while lock is not open :

keep Trying !

000 → Try 000 — work — Great,

doesn't work

001
002

Increase it by 1

using only if else condition ?

lock = 008

num = 000 , 001

[num = num+1]

num += 1

```
if (num == lock) :
    print ("Password unlocked")
else :
    num = num + 1
```

write ~8 time

→ Combination = 000

pass = 008

while (lock still closed) :

Try this combination

Combination = Combination + 1

combination != pass

001 != 008 → T

002 != 008 → T

:

False → [008 != 008]

print(combination)

Combination = 000 → 008
password = 008

001 != 008
T

{ while (combination != password) :

[combination = combination + 1] ✓

}

print (combination)

Made with Goodnotes

Numbers from 1 to 5

→ Num = 1 +1 = 2 + 1 = 3 + 1 = 4

6

2 <= 5

while Num <= 5 :

[ww print (Num)

ww num + = 1]

num = num+1

1 <= 5

4+1 = 5

6 <= 5

→ f

[1
2
3
4
5]

while + if / else

even → 7 ? 6 ?

N y

num / 2 → Remainder = 0 ?

num % 2 = = 0

start: 1 , <=10 → stop

whi

T ✓

T

✓ while num<=10 :

(if num%2==0 :

print(num)) ×

num=num+1

1 → 2

2 → 4

3 → 6 —

4 → 8 —

5 10 —

6

7

8

9

10

No of Runs of loop →

print → even No

start no is
↓
odd → increase by 1

[2, 4, 6, 8, 10]

s → 3
e → 10

4, 6, 8, 10

x

Even → Keep it as it is

s → 4
e → 10

4, 6, 8, 10

6, start = 6?

start = 6
i = start

[if start % 2 == 0 :
    i = start
  else :
    i = start + 1]

s →

6, 8

while i <= end
    print (i)
    i = i+2

1, 2, 3, 4, 5 ...

1, 3, 5, 7, 9

→ Table of 5

```
test_cases = 2
t = 1
while t <= test_cases :
  { num = 5
    i = 1
    while i <= 10 :

          print(i x num)
          i = i+1        }

    t = t+1
```

Data

[DE + BA + DA + DS + ML + AI]

(CS) — Business Analytics — Data Analytics — Science

{ Data → Decision }

Data + BA → Clean / structured data

CEO ↑ Consultant ↑ BA

[SQL + Viz + Stats] (M/M/M)

Tableau + Excel

Step 1 → SQL + Tableau + Stat → BA

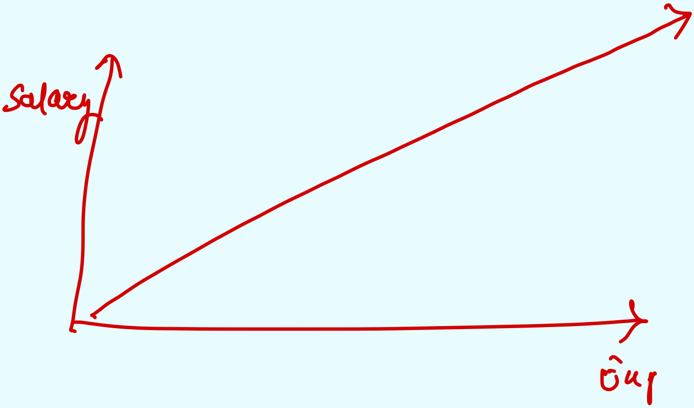
Step 2 +

unclean
Data + Adv stats → (sample)
↓
SQL + TB + Basic + Python + EDA → DA
(Product Analytics)
+
DS , ML → AI

upto 10 years :

5 yr relevent exp :-

BA → 5 x 5 → 25 LPA
DA → 5 x 6 → 30 LPA
DS → 5 x 7 → 35 LPA
DE / ML → 5 x 8 → 40 LPA
AI → 5 x 9 → 45 LPA



10 x2 = 20